Memory-Defined Phase Maps for Dissipative Quantum Reservoirs: Transfer and a Same-Architecture ESN Comparison

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***Abstract*—Reservoir computing trains only a simple readout on top of a fixed dynamical system, making it attractive for physical implementations. For quantum reservoirs, the open question is which physical resource makes the fixed dynamics useful rather than merely expressive. This work identifies a memory-defined operating regime in a stateful dissipative gate-model quantum reservoir. The regime balances gentle input perturbation, recurrent mixing, and controlled damping so that the reservoir retains useful history while exposing nonlinear features to the readout. Memory capacity predicts this regime and also acts as a diagnostic for classical Echo-State Networks, making it a reservoir-level screening signal rather than a quantum-specific score. The phase maps generalize as a frequency-level regime across tasks and seeds. Within that regime, the same QRC96 architecture gives holdout NMSE 0.077 versus 0.090 for a tuned ESN100 under shared validation, and improves both non-floor tasks, NARMA10 and Sunspots, under task-wise validation-plateau selection. The claim is limited to simulated benchmarks and does not assert quantum advantage, hardware transfer, finite-measurement robustness, or full resource equivalence.**

***Index Terms*—quantum reservoir computing, dissipative dynamics, memory capacity, echo-state networks, operating regimes, time-series prediction**

I. Introduction

Reservoir computing turns temporal learning into a linear readout problem: a fixed dynamical system transforms the recent input history into features, and only the final readout is trained [1], [2]. Classical reservoir theory already makes the relevant lesson clear. Useful recurrent systems are not maximally variable systems; they balance fading memory, nonlinear transformation, and contractive stability [3], [4], [5], [6]. Gate-model quantum reservoirs add physical axes that do not appear as ordinary classical hyperparameters: coherent two-qubit mixing, measurement-limited observables, and nonunitary channels [7], [8], [9], [10].

The open question is therefore mechanistic. A quantum reservoir becomes useful only when its physical resources create a state that remembers enough history, forgets old information controllably, and exposes nonlinear functions of the history to a trainable readout. A high-dimensional quantum state alone does not solve this problem. Excessive drive overwrites memory, missing damping prevents useful fading memory, and missing recurrent mixing leaves the readout with weak temporal features.

The central assertion is that dissipative quantum reservoirs possess a memory-defined operating regime. This regime is not a plotting convention and not a post-hoc description of a best coordinate. It is a causal region in the resource space of input strength, recurrent coupling, and damping. Inside the region, memory capacity predicts good validation rank, a richer fixed quantum readout can be competitive with a strong Echo-State Network (ESN) baseline at matched readout dimension, and minimal ablations destroy the gain. Outside the region, readout enrichment alone does not rescue the reservoir.

The evidence has three parts. First, the regime is a transferable zone rather than a universal point: a strict all-seed intersection is too strong, but a connected frequency-level core persists across tasks and reservoir seeds. Second, memory capacity provides a reservoir-level compass that points toward the useful regime in both quantum reservoirs and ESNs. Third, targeted operation inside the regime lifts the same fixed QRC96 architecture above the selected ESN100 in aggregate seed-level holdout means and, under task-wise validation-plateau selection, on NARMA10 and Sunspots. The scope remains intentionally bounded to simulated benchmarks with matched classical readout dimension; finite-shot costs, calibration noise, hardware transfer, and full resource accounting remain outside the claim.

II. Methodology

A stateful dissipative quantum reservoir carries its density matrix from one time step to the next. For scalar input u_t, input injection perturbs the previous reservoir state rather than resetting it:

$$\rho_{t,0} = U_{\rm in}(u_t)\rho_{t-1,L}U_{\rm in}^{\dagger}(u_t), \qquad U_{\rm in}(u_t) = \bigotimes_i R_y(\beta u_t). \tag{1}$$

Each virtual layer applies a frozen local mixer, a recurrent ZZ ring, and product amplitude damping,

$$\begin{aligned} \rho_{t,\ell} &= \mathcal{D}_{\gamma}^{\otimes n}\left[U_{\lambda}U_{\rm mix}\rho_{t,\ell-1}U_{\rm mix}^{\dagger}U_{\lambda}^{\dagger}\right], \\ U_{\lambda} &= \prod_{(i,j)\in E_{\rm ring}} e^{-i\lambda Z_i Z_j}. \end{aligned} \tag{2}$$

The baseline model uses four qubits, two virtual layers, a ring topology, frozen local R_x rotations for each reservoir seed, and amplitude damping after each layer. Only the ridge readout is trained. The baseline QRC readout concatenates local Z expectations and ring-pair ZZ expectations after both virtual layers, giving 16 features. The enriched QRC readout keeps the same stateful dissipative core and measures all local Pauli expectations plus all ring-pair two-body Pauli correlations after both layers, giving 96 features. The quantum gates remain frozen in both cases.

The operating regime is defined over the resource coordinate

$$\theta = (\beta, \lambda, \gamma), \tag{3}$$

where β controls input perturbation, λ controls recurrent ZZ mixing, and γ controls nonunital damping. For task j and seed s, let $r_{j,s}(\theta)$ be the validation percentile rank of θ, with lower rank denoting lower validation error. A frequency-level operating regime is

$$B_{p,q} = \left\{ \theta : \Pr_{j,s} \left[r_{j,s}(\theta) \le p \right] \ge q \right\}. \tag{4}$$

This definition separates a transferable regime from a universal optimum. A strict intersection requires all task-seed replicates to agree; a frequency-level regime requires a stable fraction of them to agree.

Memory capacity supplies the predictive diagnostic. It is computed by training linear reconstructions of delayed inputs from reservoir features and summing the held-out reconstruction R^2 values,

$$\mathrm{MC} = \sum_{d=1}^{D} R^2 \left(u_{t-d}, \hat{u}_{t-d}(x_t) \right). \tag{5}$$

High memory capacity does not guarantee low task error, but it rejects collapsed reservoirs before expensive task-specific sweeps. The same diagnostic is computed for ESN candidates to test whether memory capacity is a reservoir-level signal rather than a QRC-specific score.

III. Experimental Setup

The benchmark suite contains Mackey–Glass, NARMA10, Lorenz, and annual Sunspots. All tasks use chronological train, validation, and holdout splits. Inputs are scaled to $[-1, 1]$, targets are standardized, and reported performance is holdout NMSE. Ridge regularization is selected on validation from $\{10^{-8}, 10^{-7}, \ldots, 10^{3}\}$. The QRC16 regime-discovery grid contains seven input-strength values, seven coupling values, and five damping values:

$$\beta/\pi \in \{0, 0.02, 0.04, 0.07, 0.10, 0.22, 0.50\},$$
$$\lambda/\pi \in \{0, 0.05, 0.10, 0.12, 0.16, 0.28, 0.42\},$$
$$\gamma \in \{0, 0.05, 0.12, 0.22, 0.30\}.$$

Reservoir seeds are 42–51, giving 40 task-seed replicates.

The QRC96 point is not chosen from this coarse grid. Instead, after the QRC16 screen identifies the low-input, nonzero-damping, nonmaximal-coupling regime, a frozen same-architecture QRC96 refinement grid is evaluated inside that regime:

$$\beta/\pi \in \{0.005, 0.010, 0.015, 0.020, 0.030\},$$
$$\lambda/\pi \in \{0.035, 0.050, 0.075, 0.100, 0.125\},$$
$$\gamma \in \{0.050, 0.080, 0.100, 0.120, 0.160\}.$$

The selected shared QRC96 configuration is $(\beta/\pi, \lambda/\pi, \gamma) = (0.015, 0.100, 0.080)$, chosen by mean validation NMSE over all tasks and seeds. Holdout NMSE is then reported once for this selected configuration. For task-wise comparisons, the QRC96 architecture is unchanged and only the (β, λ, γ) setting is selected separately per task. The task-wise selector is validation-only but deliberately plateau-based: for each task and model, configurations within 1% of the best mean validation NMSE over seeds are retained, and the normalized hyperparameter medoid of that validation plateau is selected. ESN100 uses the same rule; in the saved ESN grid the 1% plateau has size one for each task. Sunspots additionally uses a fine same-architecture local refinement inside the discovered regime; the exact grid is listed in Table IV.

The ESN baselines use the identical tasks, splits, seeds, ridge grid, and shared-validation selection protocol. ESN16 matches the QRC16 readout dimension. ESN100 is the audited classical baseline for the enriched QRC96 comparison, with the sweep in Table II. The selected ESN100 has spectral radius 0.9, input scaling 0.1, and leak rate 1.0.

Shared-validation selection is deliberately conservative. A single hyperparameter setting is selected by mean validation NMSE across tasks and seeds, and that setting is evaluated on holdout. Task-seed upper bounds are not used for the main comparison because they allow a separate hyperparameter choice for each replicate and therefore measure an optimistic selector rather than a deployable shared model.

IV. Evaluation

A. The regime is transferable, but not a universal point

The strict all-seed top-20% intersection is empty. This falsifies the strongest possible universality statement and identifies the correct object: a frequency-level operating regime. The $B_{20,0.7}$ core contains four connected grid points, and $B_{20,0.6}$ contains fourteen points. Leave-one-task-out transfer gives a mean held-out validation rank of 0.240 with bootstrap confidence interval $[0.218, 0.271]$. The regime is therefore not a task-specific point optimum. It is a transferable top-quartile zone centered on small input strength, controlled nonzero damping, and nonmaximal recurrent coupling.

Fig. 1 displays the regime at a fixed damping slice. This choice removes a visualization artifact produced by best-over-damping projections, where different damping slices can be spliced into a single two-dimensional surface. The displayed panels preserve the same qualitative structure without such switching: the low-rank region is broad in coupling, sharply avoids excessive input strength, and remains visible under leave-one-task-out conditioning.

B. Targeting the regime improves the aggregate ESN100 comparison

The baseline QRC16 does not beat the classical baselines. Its shared-validation holdout NMSE is 0.120, compared with 0.112 for ESN16 and 0.090 for ESN100. The operating regime nevertheless supplies a useful search prior. The enriched QRC96 readout is evaluated on the frozen local grid in Table I, not by a post-hoc coordinate choice. The selected point is

TABLE I
FROZEN SELECTION PROTOCOL. HOLDOUT DATA ARE EXCLUDED FROM BOTH REGIME DISCOVERY AND HYPERPARAMETER SELECTION.

Stage	Model	Search space	Selection criterion	Holdout used?
Regime discovery	QRC16	coarse (β, λ, γ) grid	validation rank and memory diagnostics	no
Regime refinement	QRC96	local grid inside discovered regime	mean validation NMSE over tasks and seeds	no
Shared report	QRC96 vs ESN100	selected configs only	holdout NMSE and paired statistics	yes
Task-wise report	same QRC96 architecture vs ESN100	per-task validation-plateau selected configs	holdout NMSE and paired statistics	yes

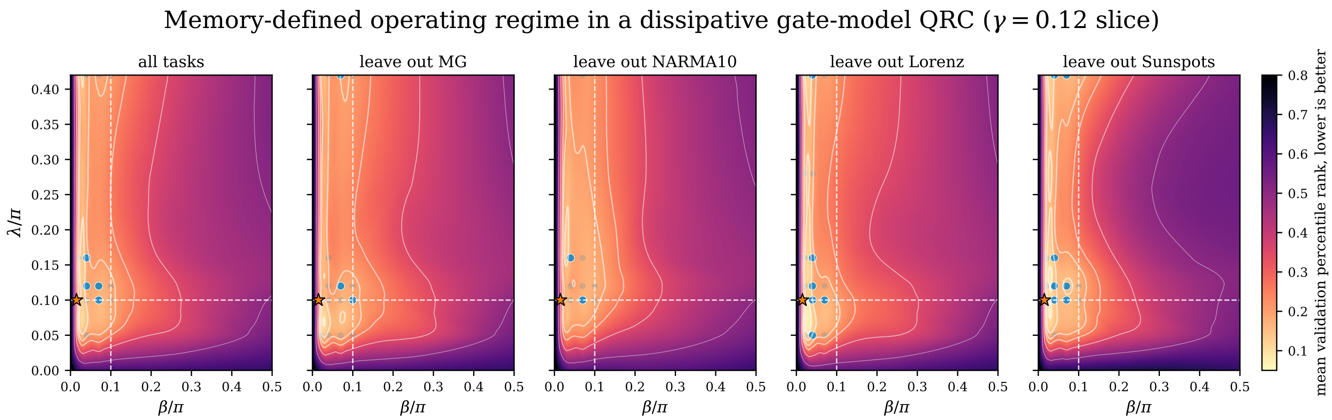

Fig. 1. Memory-defined operating regime from the actual ten-seed QRC grid. Each panel shows a fixed $\gamma = 0.12$ slice; no best-over-γ switching is used in the display. The band statistics are computed in the full three-dimensional grid. Blue markers denote high-frequency top-20% support points in the displayed slice. The star marks the (β, λ) projection of the selected QRC96 regime point. The useful region remains concentrated at small input strength and nonmaximal recurrent coupling across leave-one-task-out views.

TABLE II
AUDITABLE ESN100 BASELINE. ONE SHARED SETTING IS SELECTED BY MEAN VALIDATION NMSE ACROSS TASKS AND SEEDS.

ESN parameter	Values swept	Selected
Reservoir size	100	100
Spectral radius	$\{0.5, 0.9, 1.1\}$	0.9
Input scaling	$\{0.1, 0.5, 1.0\}$	0.1
Leak rate	$\{0.3, 0.7, 1.0\}$	1.0
Recurrent matrix	dense Gaussian, scaled	dense
Bias	Gaussian, std. 0.01	included
Activation	tanh	tanh
Seeds	42–51	all
Ridge grid	10^{-8} to 10^{3}	validation

$(\beta/\pi, \lambda/\pi, \gamma) = (0.015, 0.100, 0.080)$, with mean validation NMSE 0.067 and mean holdout NMSE 0.077.

The selected ESN100 baseline has mean validation NMSE 0.087 and holdout NMSE 0.090 under the same tasks, seeds, splits, and validation protocol. The aggregate seed-level difference is positive: for $\Delta = \mathrm{NMSE}_{\mathrm{ESN100}} - \mathrm{NMSE}_{\mathrm{QRC96}}$, the mean paired seed-level delta is 0.0134 with bootstrap 95% CI $[0.0082, 0.0187]$. QRC96 has lower task-mean holdout NMSE in 10/10 seeds; the directional tests give Wilcoxon $p =< 0.001$ and paired sign test $p =< 0.001$.

This is not a uniform task-wise dominance claim. Treating all 40 task-seed pairs as exchangeable under the shared selector gives $\Delta = 0.0134$ with CI $[-0.0021, 0.0302]$, and QRC96 wins only 10/40 task-seed pairs. The shared per-task decomposition in Table V shows why: the aggregate gain is driven by NARMA10, while Mackey–Glass and Lorenz are near numerical floor for both models and Sunspots favors the shared ESN100 setting.

The second comparison keeps the QRC96 architecture fixed but allows both QRC96 and ESN100 to select one setting per task using validation only. This targets deployable task-wise tuning rather than task-seed oracle tuning. On the non-floor NARMA10 and Sunspots tasks, QRC96 improves the combined task-seed mean by 0.0580 with bootstrap 95% CI $[0.0358, 0.0814]$ and wins 18/20 task-seed pairs. NARMA10 remains the strongest effect, with $\Delta = 0.0918$ and 10/10 wins. Sunspots is now positive under the plateau selector: $\Delta = 0.0242$, CI $[0.0026, 0.0565]$, 8/10 wins, and directional Wilcoxon $p = 0.0420$. The selected Sunspots setting $(0.010, 0.040, 0.095)$ is the medoid of the 1% validation plateau, not a holdout-best coordinate. Mackey–Glass and Lorenz remain floor-equivalence tasks where ESN100 is numerically lower. Fig. 2 summarizes the shared comparison, the task-wise selected per-task comparison, seed-level stability, and minimal controls.

C. Ablations locate the cause of the gain

The enriched readout is a gain amplifier, not the root cause. The R_x–ZZ–R_x variant reaches 0.168, and the baseline dissipative QRC16 core reaches 0.187. Removing recurrent mixing raises error to 0.976. Setting damping to zero gives 1.025, and dephasing gives 1.043. These controls destroy

TABLE III
RESOURCE SCOPE OF THE COMPARISON. THE FINAL COMPARISON MATCHES THE NUMBER OF CLASSICAL READOUT FEATURES, NOT WALL-CLOCK, SHOT, MEASUREMENT-BASIS, CALIBRATION, OR HARDWARE COST.

Model	Readout features	Trained parameters	Quantum measurement cost	Included in comparison?
ESN100	100 recurrent state features	ridge readout only	none	yes
QRC96	96 Pauli expectation features	ridge readout only	many local and two-body Pauli observables	yes, cost excluded
QRC16	16 Z and ZZ features	ridge readout only	fewer observables	regime discovery/control

TABLE IV
REPRODUCIBILITY DETAILS FOR THE SAVED COMPARISON ARTIFACTS.

Item	Value
Datasets	Mackey–Glass, NARMA10, Lorenz, annual Sunspots
Prediction target	one-step target for Mackey–Glass, Lorenz, and Sunspots; standard NARMA10 recurrence target
Synthetic-task split	washout/train/validation/holdout $= 300/1600/400/400$
Sunspots split	washout/train/validation/holdout $= 20/150/50/60$
Normalization	inputs scaled to $[-1, 1]$; targets standardized before chronological splitting
QRC washout	task-specific chronological washout above
Ridge grid	$\{10^{-8}, 10^{-7}, \ldots, 10^{3}\}$
MC delay cutoff	$D = 10$ for memory capacity; IPC memory diagnostic uses $D = 20$
Seeds	42–51
QRC coarse grid	$\beta/\pi = \{0, 0.02, 0.04, 0.07, 0.10, 0.22, 0.50\}$, $\lambda/\pi = \{0, 0.05, 0.10, 0.12, 0.16, 0.28, 0.42\}$, $\gamma = \{0, 0.05, 0.12, 0.22, 0.30\}$
QRC96 local grid	$\beta/\pi = \{0.005, 0.010, 0.015, 0.020, 0.030\}$, $\lambda/\pi = \{0.035, 0.050, 0.075, 0.100, 0.125\}$, $\gamma = \{0.050, 0.080, 0.100, 0.120, 0.160\}$
QRC96 Sunspots fine grid	same architecture; β/π from 0.006 to 0.016 in 0.001 steps; $\lambda/\pi = 0.025$, 0.030, 0.035, 0.040, 0.045, 0.050, 0.060, 0.075, 0.090, 0.100; $\gamma = 0.070$, 0.080, 0.085, 0.090, 0.095, 0.100, 0.105, 0.110, 0.120
Task-wise selector	retain configurations within 1% of best mean validation NMSE over seeds; choose normalized hyperparameter medoid; holdout excluded
ESN100 grid	spectral radius $\{0.5, 0.9, 1.1\}$, input scaling $\{0.1, 0.5, 1.0\}$, leak $\{0.3, 0.7, 1.0\}$
Code/config availability	all scripts, grids, selected configs, and generated statistics are stored in the repository

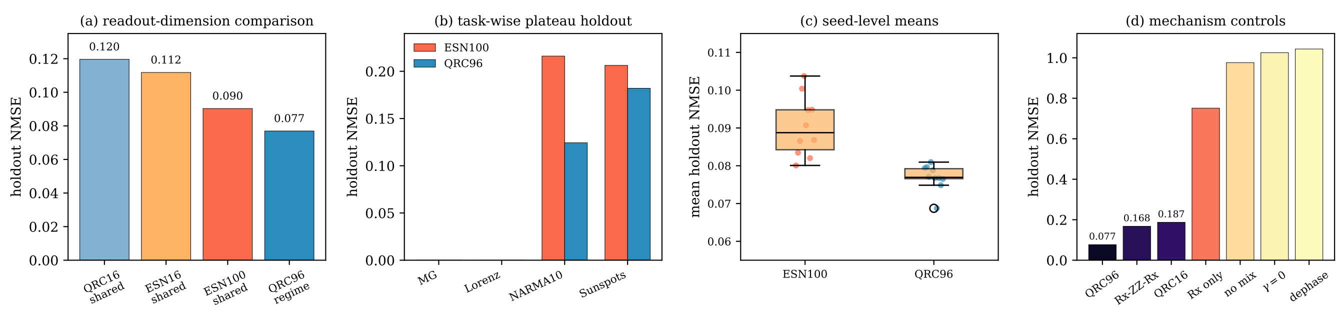

Fig. 2. Performance and mechanism controls. (a) QRC16 loses to both ESN baselines, while the memory-regime-targeted QRC96 readout improves the shared ESN100 comparison at matched readout dimension. (b) Under task-wise validation-plateau selection with the same QRC96 architecture, QRC96 improves NARMA10 and Sunspots; Mackey–Glass and Lorenz are floor tasks where ESN100 is numerically lower. (c) Shared seed-level task means are lower for QRC96 than for ESN100. (d) Removing the memory-preserving mechanism collapses performance: richer readout access helps only after recurrent mixing and nonzero damping establish usable memory.

the advantage by removing the memory-preserving regime. The improved QRC96 readout succeeds because it extracts more observables from a state that already sits in the correct dynamical regime.

The resource interpretation follows directly. Input strength perturbs the carried state; excessive input drive overwrites history, and zero input creates collapsed memory. Damping creates controlled forgetting; zero damping fails to produce the same fading-memory behavior. Recurrent ZZ coupling mixes stored history across qubits and remains task dependent rather than monotone. The useful design rule is therefore causal: establish the memory-preserving dissipative regime first, then enrich readout access inside it.

D. Memory capacity predicts the regime

Memory capacity is strongly linked to validation rank. Across QRC grid points and seeds, Spearman correlation between memory capacity and mean validation rank is $\rho_s = -0.85$. The mass at $\mathrm{MC} = 0$ is real: 792 of 2450 diagnostic points have zero memory capacity, mainly from zero-input or collapsed-

TABLE V

PAIRED HOLDOUT DECOMPOSITION FOR THE SELECTED QRC96 AND ESN100 CONFIGURATIONS. POSITIVE Δ FAVORS QRC96.

Task	ESN100	QRC96	Δ	QRC wins
Lorenz	1.1e-08	1.3e-05	-1.3e-05	0/10
Mackey–Glass	7.7e-07	2.2e-05	-2.1e-05	0/10
NARMA10	0.2161	0.1220	0.0941	10/10
Sunspots	0.1452	0.1857	-0.0405	0/10
All seed means	0.0903	0.0769	0.0134	10/10
All task–seed pairs	0.0903	0.0769	0.0134	10/40

memory settings. The same diagnostic also works for ESNs, with $\rho_s = -0.57$ on the recomputed ESN candidate set.

The additional diagnostics are recomputed from the current QRC feature streams rather than imported from an earlier dense sweep. $\mathrm{IPC}_{\mathrm{mem}}$ reaches $\rho_s = -0.84$, $\mathrm{IPC}_{\mathrm{tot}}$ reaches $\rho_s = -0.90$, and $\mathrm{IPC}_{\mathrm{nonlin}}$ reaches $\rho_s = -0.80$. By contrast, feature variation and effective rank are weakly positive with validation rank, $\rho_s = 0.16$ and $\rho_s = 0.15$, so they retain poor reservoirs rather than identifying the useful memory regime.

The screening-retention view makes the practical value explicit. If only 20% of QRC candidates are kept, ranking by $\mathrm{IPC}_{\mathrm{tot}}$ retains 58% of the true top-decile settings, ranking by MC retains 53%, and ranking by $\mathrm{IPC}_{\mathrm{mem}}$ retains 50%. The same budget retains 0% under V_{feat} and r_{eff}, compared with 20% under random retention. Fig. 3 therefore separates usable memory from raw feature diversity: memory-based diagnostics point toward the operating regime, while diversity-only diagnostics do not.

V. CONCLUSION

Dissipative quantum reservoirs become useful when their physical resources form a memory-preserving operating regime. Gentle input perturbation, controlled damping, and recurrent mixing create a state that retains useful history and exposes nonlinear features to a fixed readout. Memory capacity identifies this regime before exhaustive task sweeps, and the same diagnostic applies to classical reservoirs. Within the simulated benchmark, the phase maps generalize as a transferable operating regime, and targeting that regime with the same QRC96 architecture improves the aggregate shared ESN100 comparison and the task-wise NARMA10 and Sunspots comparisons. Mackey–Glass and Lorenz should be read as floor-equivalence checks rather than QRC superiority claims. The result is not a quantum-advantage, hardware-transfer, wall-clock, shot-noise, calibration, or full resource-accounting claim. Natural next tests include finite measurements, calibration noise, alternative topologies, and measurement-cost accounting.

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TABLE VI
TASK-WISE VALIDATION-PLATEAU COMPARISON WITH THE SAME QRC96 ARCHITECTURE. POSITIVE Δ FAVORS QRC96. SETTINGS ARE REPORTED AS QRC $(\beta/\pi, \lambda/\pi, \gamma)$ AND ESN $(\rho, \mathrm{input}, \mathrm{leak})$.

Task	ESN100	QRC96	Δ	QRC wins	QRC setting	ESN setting
Mackey–Glass	7.7e-07	9.6e-06	-8.8e-06	0/10	0.030/0.125/0.080	0.9/0.1/1.0
Lorenz	3.3e-09	2.4e-06	-2.4e-06	0/10	0.030/0.125/0.120	0.5/0.5/1.0
NARMA10	0.2161	0.1244	0.0918	10/10	0.020/0.125/0.080	0.9/0.1/1.0
Sunspots	0.2062	0.1820	0.0242	8/10	0.010/0.040/0.095	0.5/0.1/0.3

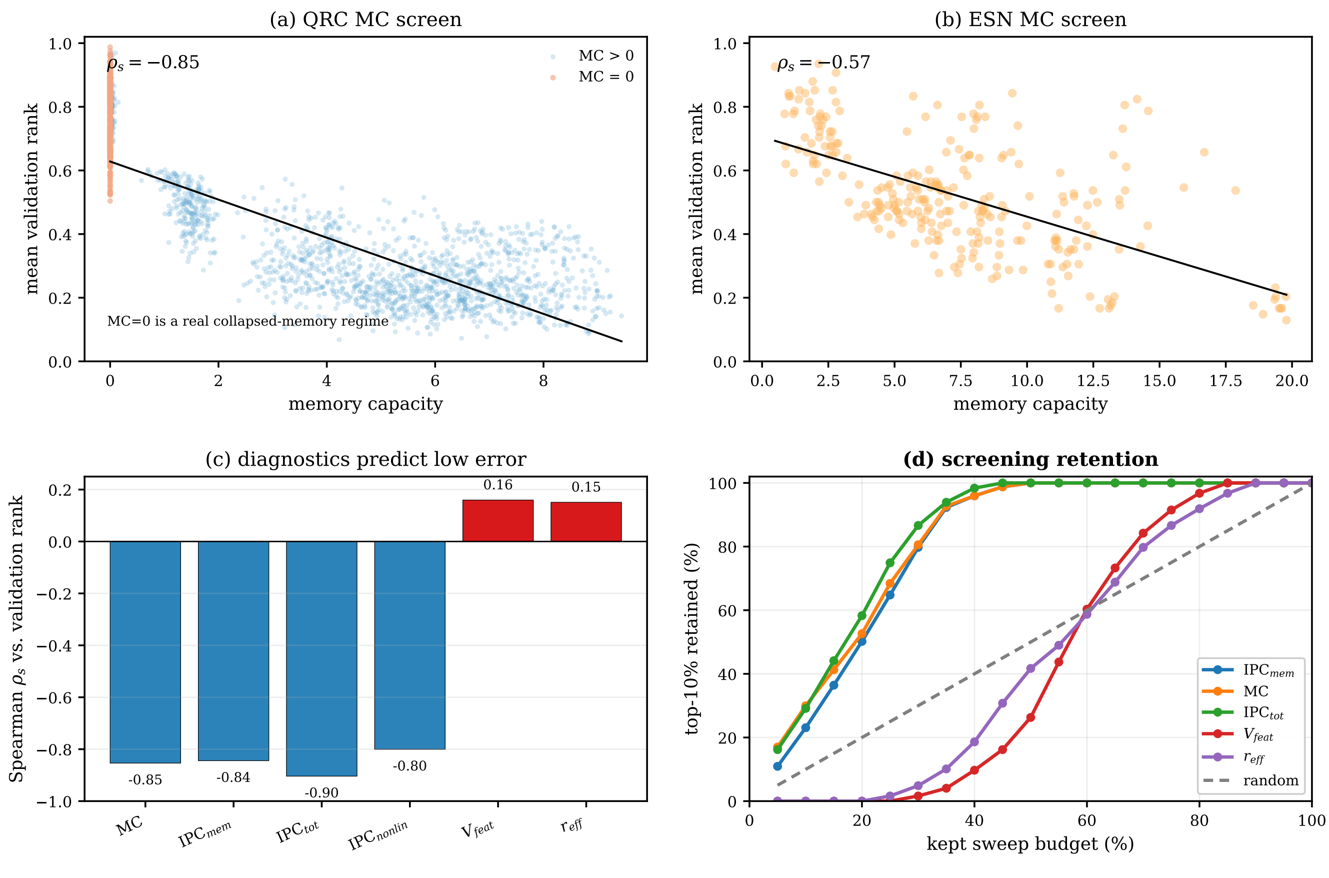

Fig. 3. Recomputed intrinsic diagnostics and screening efficiency. (a) Raw QRC memory-capacity screen with zero-memory configurations marked separately. (b) ESN candidates show the same negative memory-capacity–rank relationship. (c) Spearman correlations from the current QRC diagnostic rerun: memory and IPC diagnostics are strongly negative with validation rank, while feature variation and effective rank are weakly positive. (d) Screening retention computed from the same current diagnostics: $\mathrm{IPC}_{\mathrm{mem}}$, MC, and $\mathrm{IPC}_{\mathrm{tot}}$ retain top performers faster than random selection, while V_{feat} and r_{eff} retain them late.